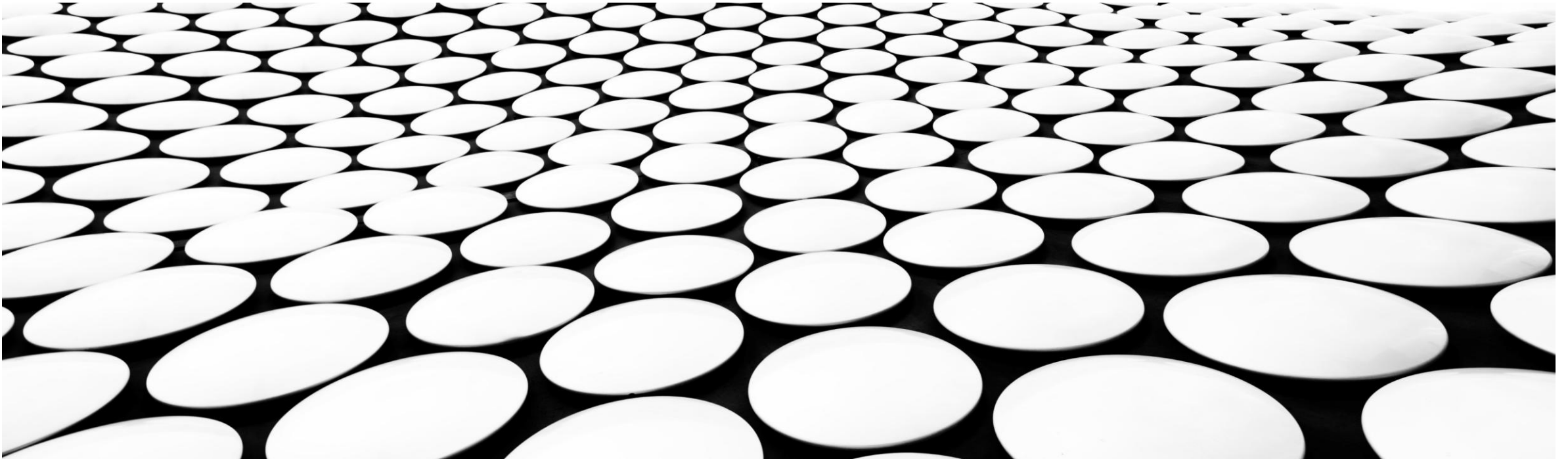
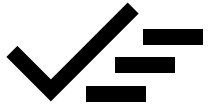


DATA SCIENCE PROCESS

Summary case study





DATA SCIENCE PROCESS

- Introduction
- Case Study – Streaming Platform
 - Interaction with the manager
 - Interaction with the data engineer
 - Interaction with the UI designer
 - Interaction with the ML expert
- Summary

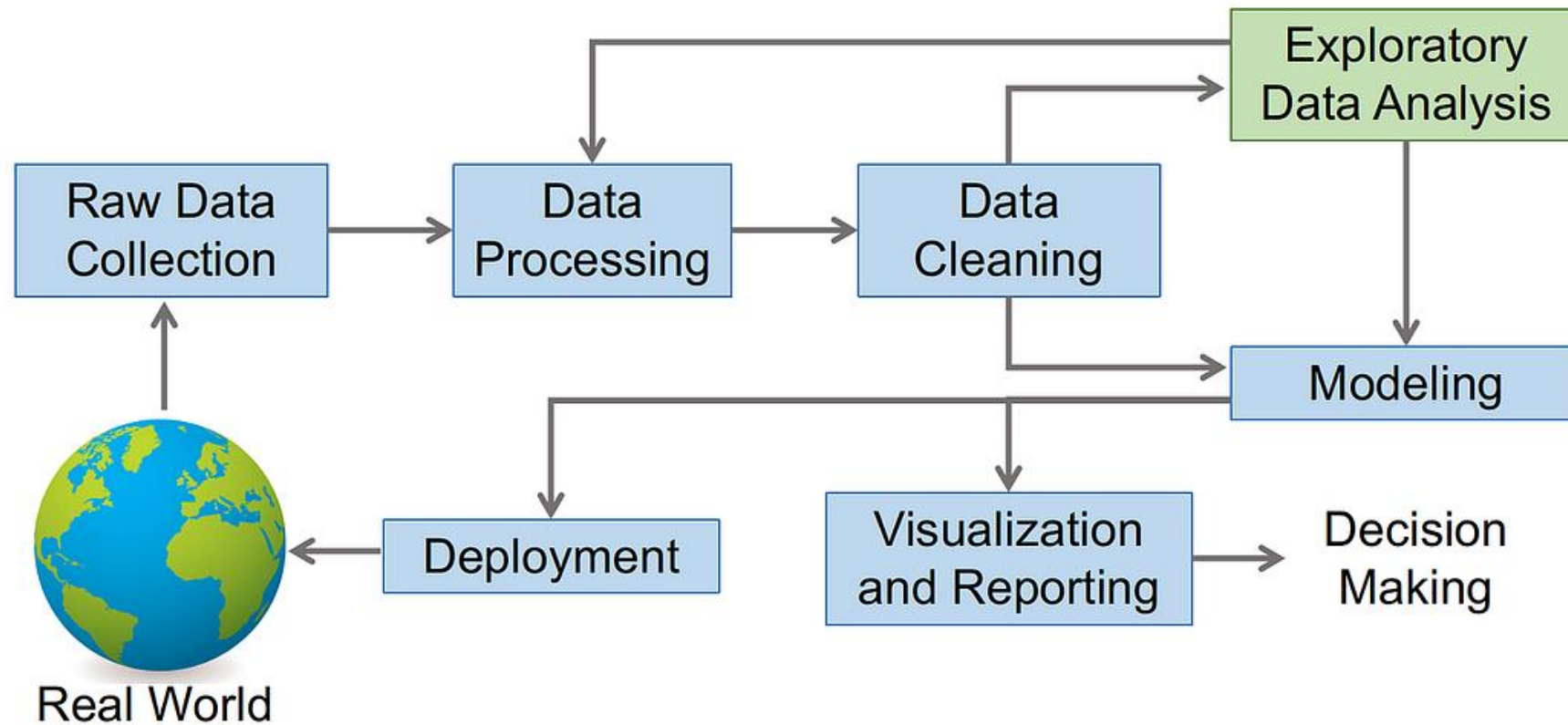
Introduction

Topics covered

What we saw this semester:

- Week 1: Insights and Data
- Week 2: Exploratory Data Analysis
- Week 3: Data collection methods
- Week 4: Data Cleaning
- Week 6: Missing Data and Biases
- Week 7: Predictive Modeling (Classification and Regression)
- Week 8: Feature engineering and Ensemble models
- Week 9: Clustering and Recommendation
- Week 11: Time series
- Week 12: Data storage and integration

Data Science Process



Skills

To begin with Data Science Modelling, the Data Science Companies expect the ideal candidate to know some skills.



Source

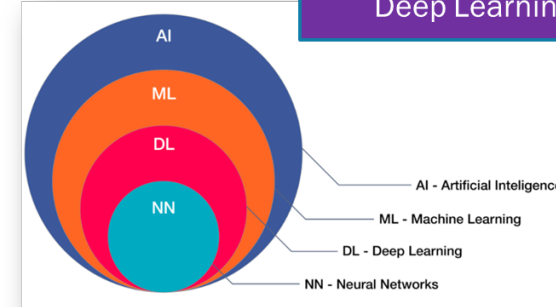
Statistics and Probability



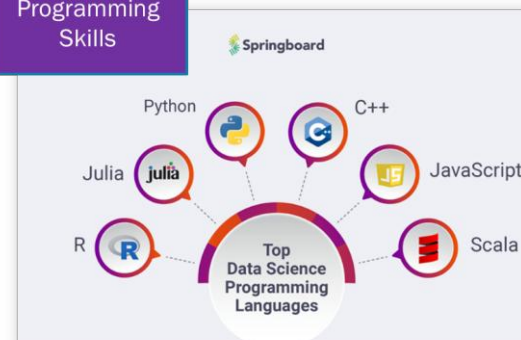
Data Visualization Skills



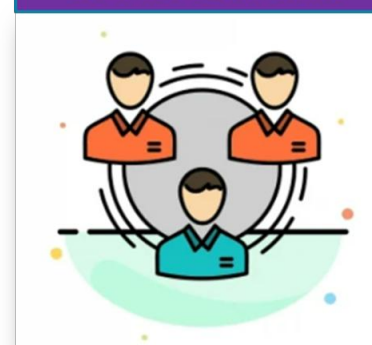
Machine Learning and Deep Learning



Programming Skills

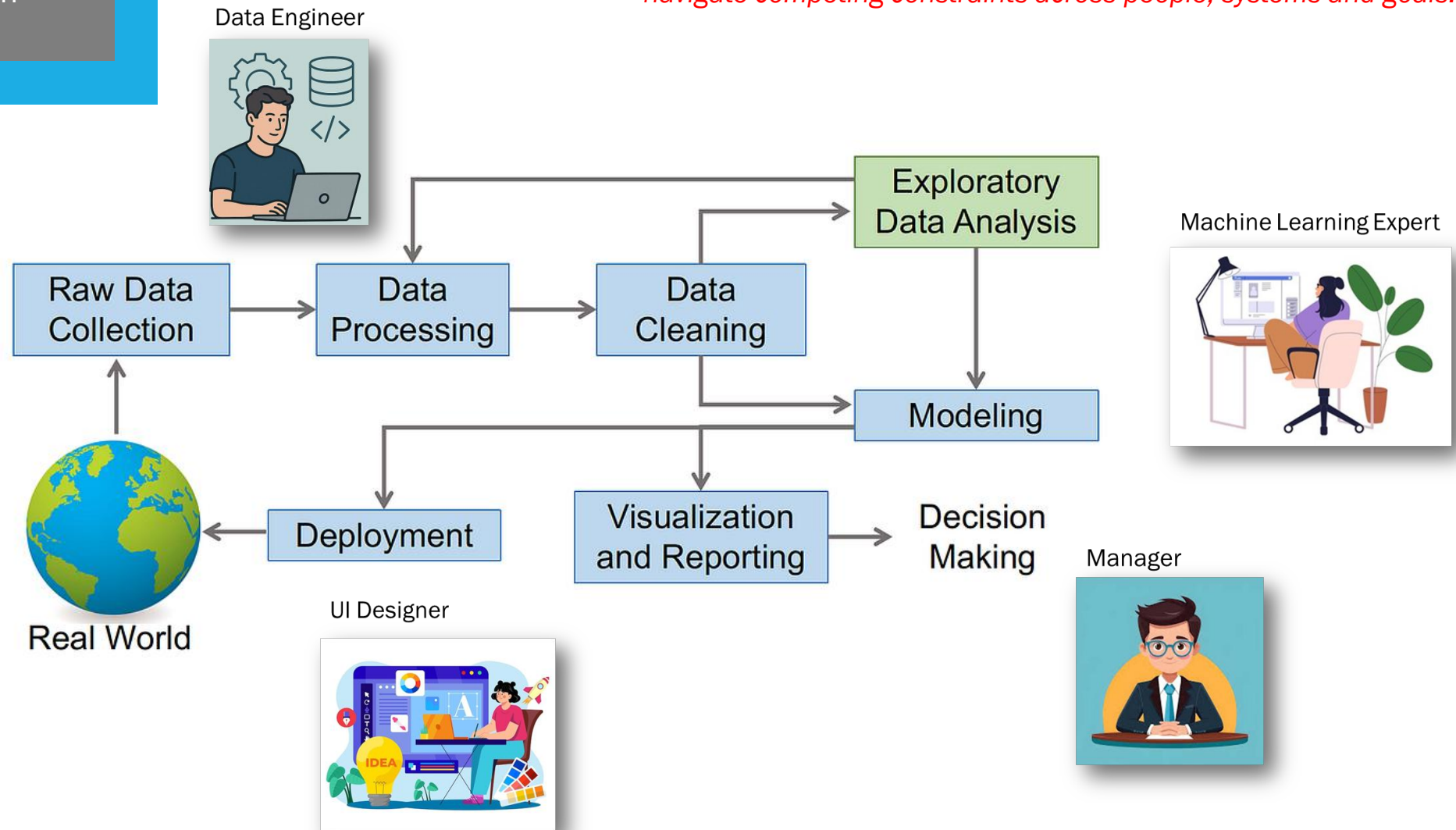


Communication Skills



Data Ecosystem

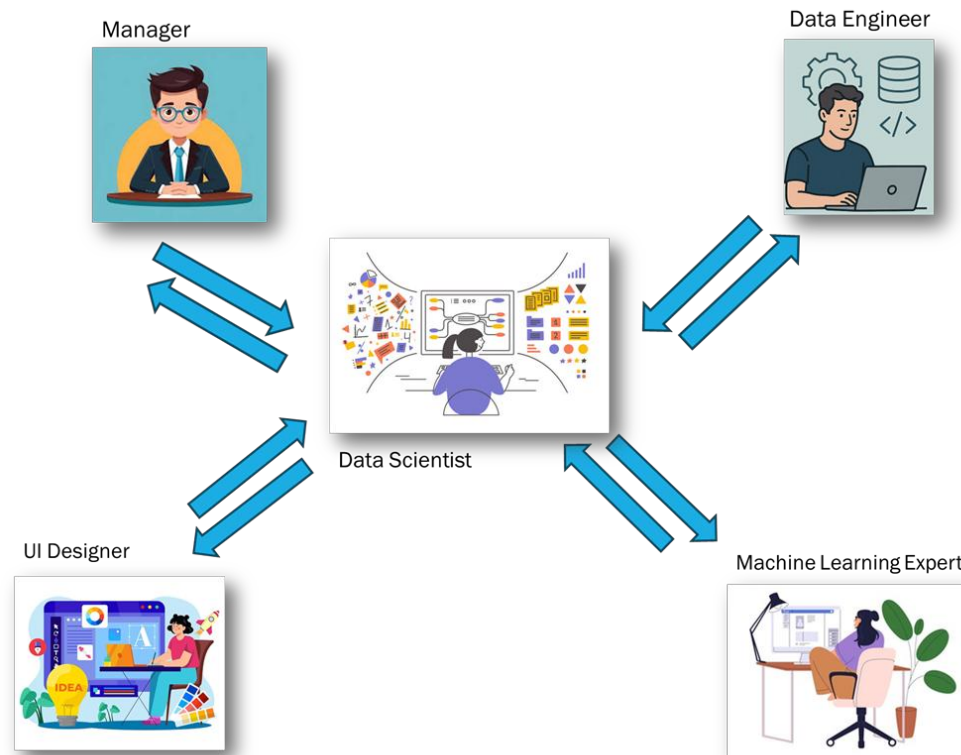
A data scientist job is to well understand the data process and also navigate competing constraints across people, systems and goals.



Today's approach

- Synthesize everything learned
- Position the Data Scientist at the center of an ecosystem of experts
- Understand communication differences
- Understand goal differences

Case Study: Movie Streaming platform

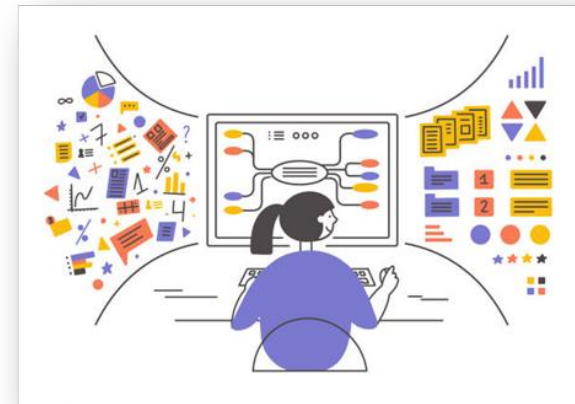
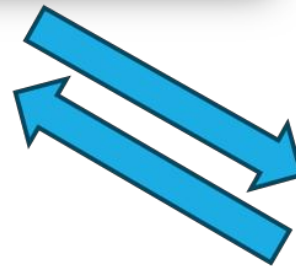


The manager



- Subscriber growth and retention
- Watch time
- Revenue impact
- Clear, actionable insights

Manager



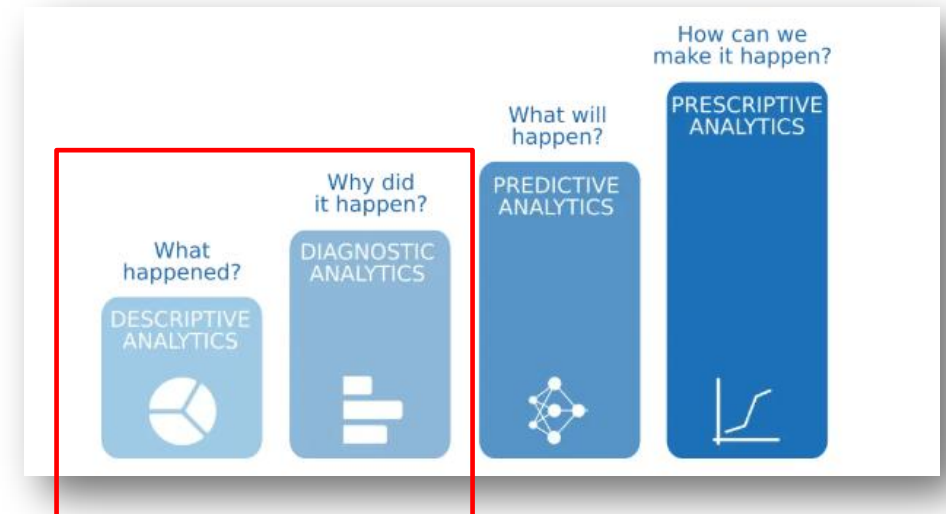
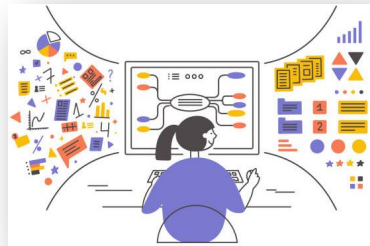
Data Scientist

Manager



We hired you to figure out why users are leaving the platform.
We want you to help us find out more about our viewers.

OK, I'll start by generating descriptive statistics on the viewers. Then we can explore correlations between their characteristics and leaving.



Toolbox

Descriptive Statistics

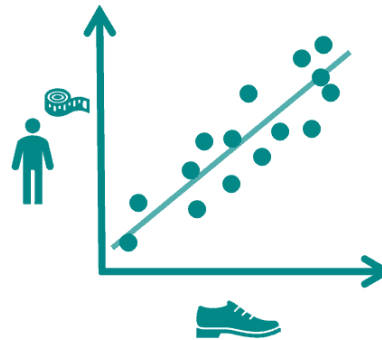
Measures of central tendency

1. Mean
2. Median
3. Mode

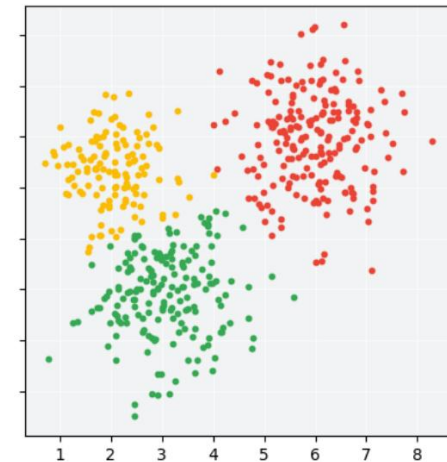
Measures of dispersion

1. Range
2. Variance
3. Standard deviation

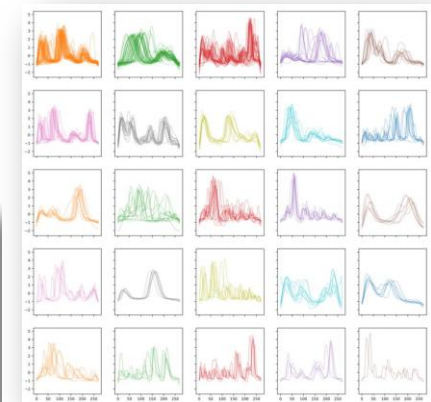
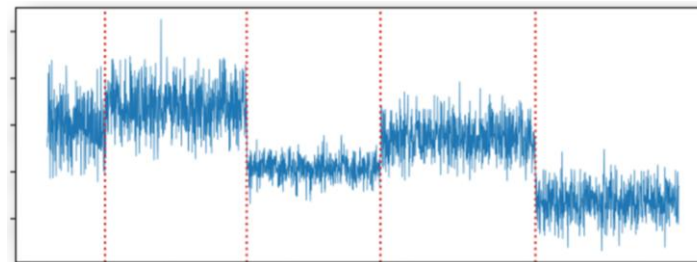
Diagnostic Statistics



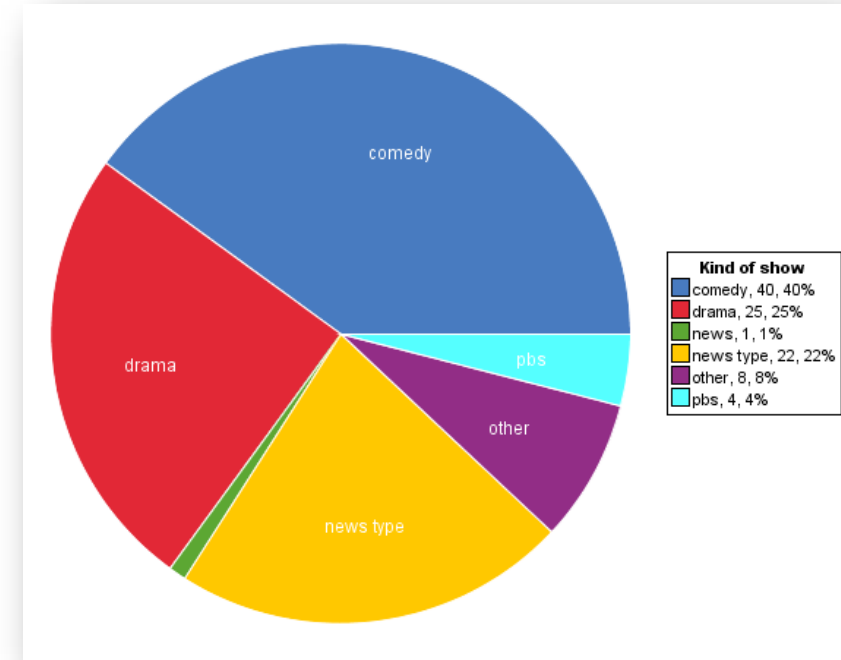
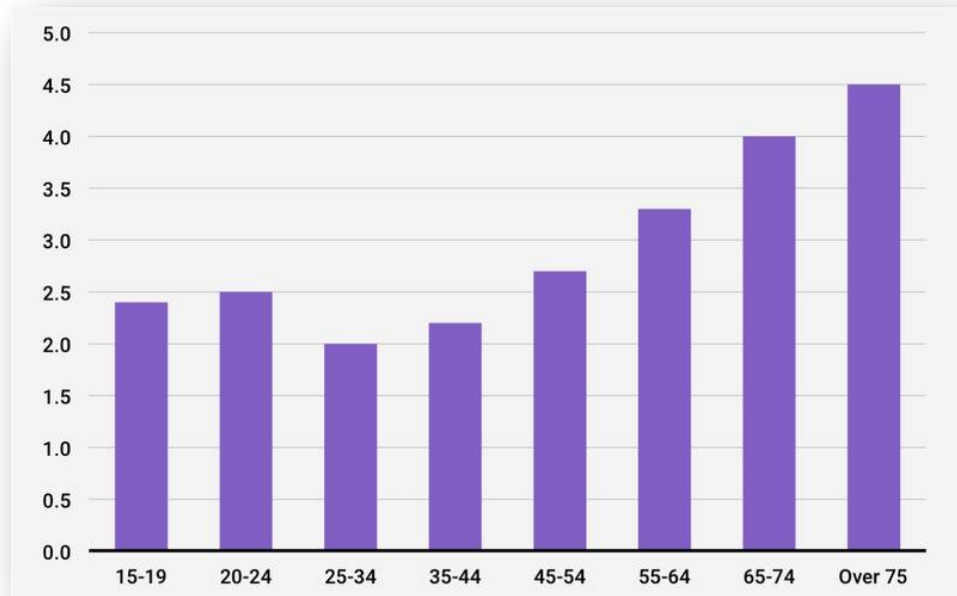
Clustering approaches



Time Series analysis



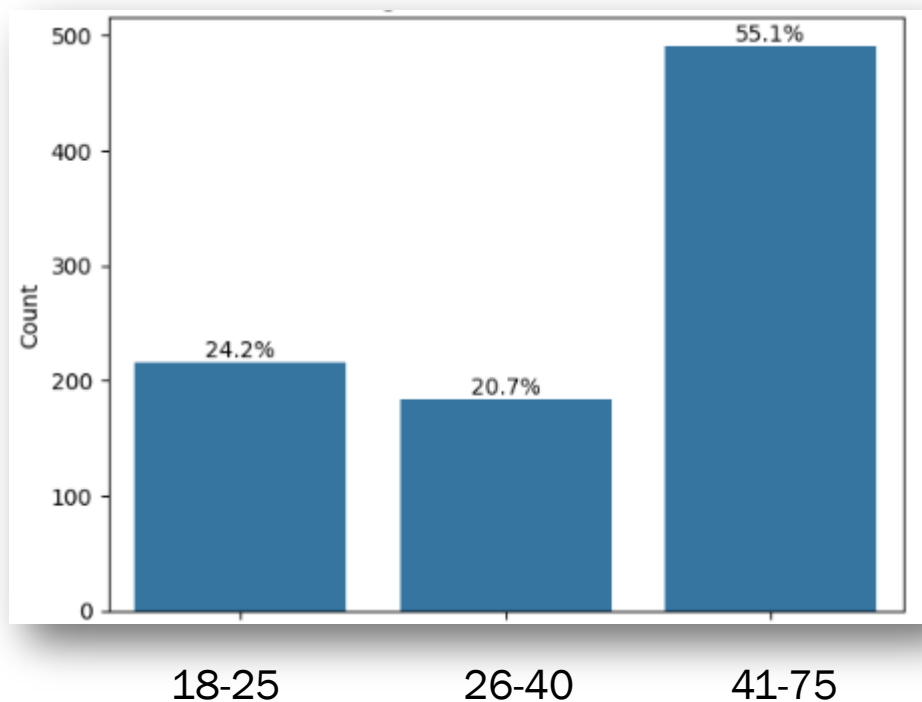
Descriptive Statistics



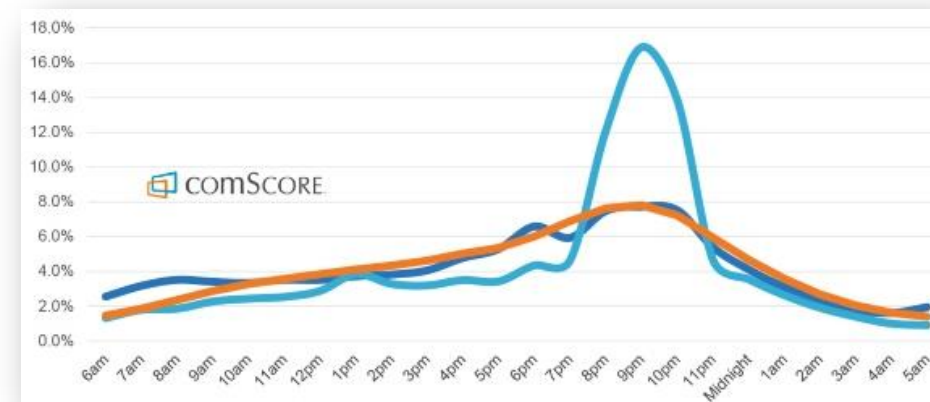
You decide to show statistics with particular groupings

- Numerical/Numerical
- Numerical/Categorical
- Categorical/Categorical

Churn rate (after the one-month free trial)



Time Series analysis



Different age groups

We'll have to think about these bins



Data Sources

Data



Data Science
Process



Insight

Ideally we would have:

User metadata

- Age
- Gender
- Address

Content Metadata

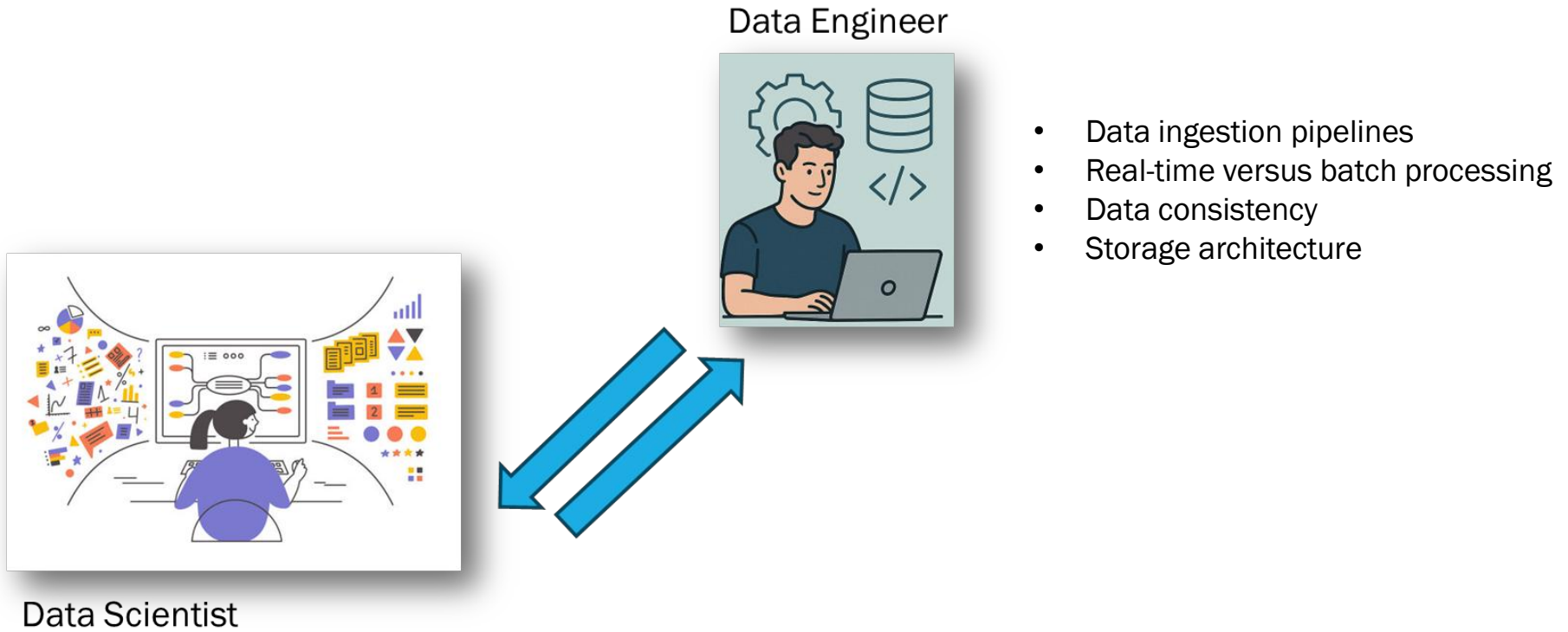
- Movie title
- Actors
- Genre

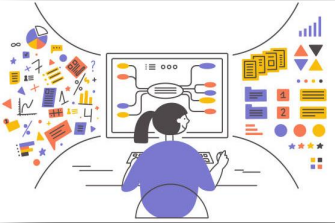
Viewing history

- What users watched
- Duration per session
- Ratings and feedback

The data engineer





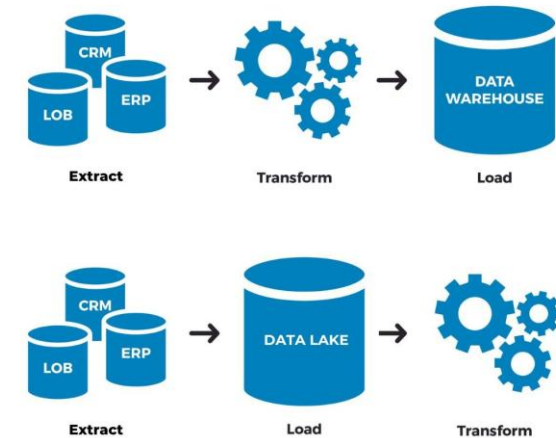


Can you tell me about data acquisition and storage?
What kind of pipeline do you use?

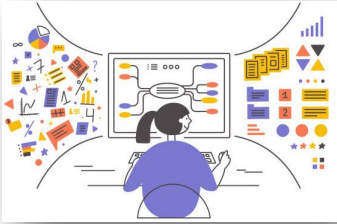
We gather lots of data and we just keep everything in a data lake now... we're not yet focused on specific types of analysis.



ETL
VS.
ELT

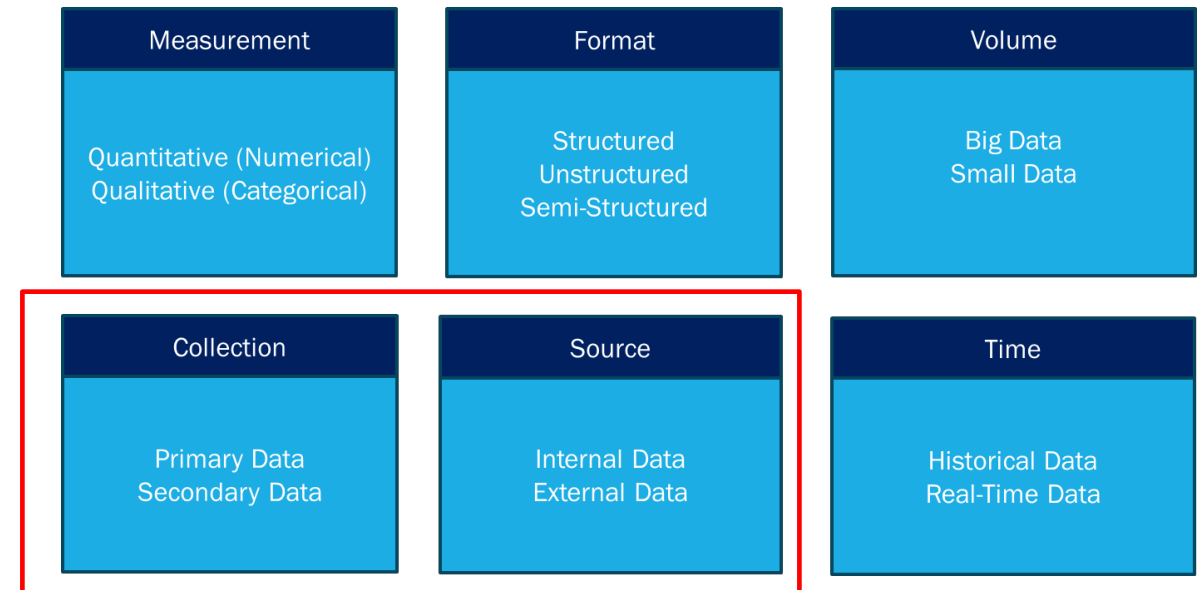


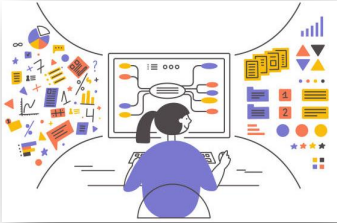
bismart



Is it primary internal data? Any external source used?

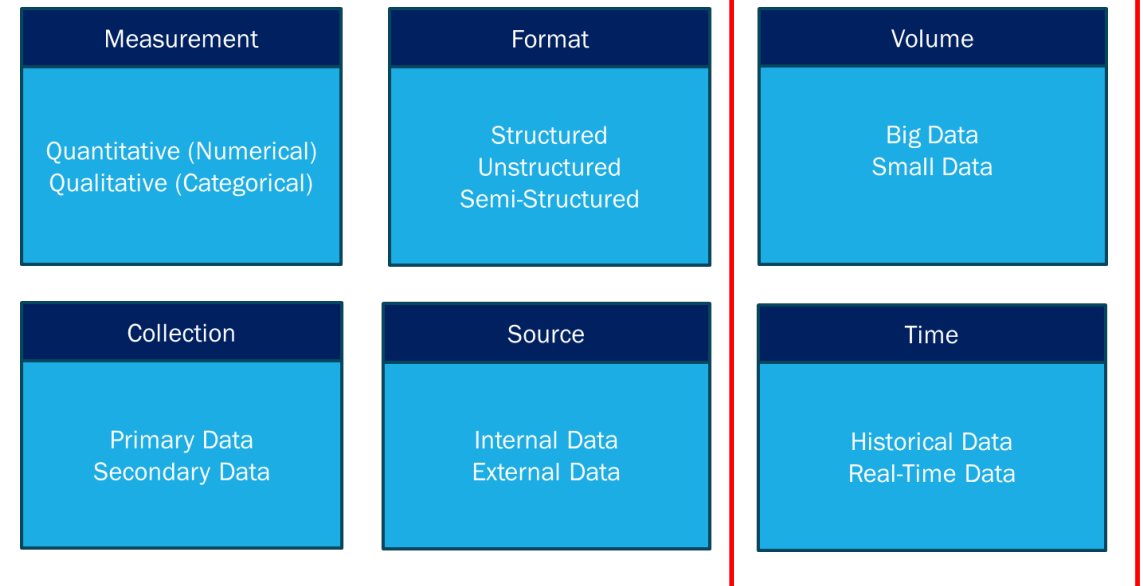
We have internal data for customer info and their viewing logs, but external for the movie metadata.

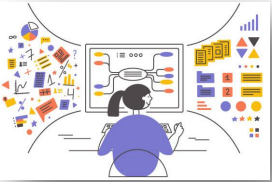




Viewing logs? That's probably lots of data.
What is your retention period?

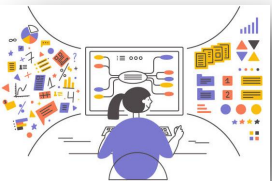
One year. And if the number of users increase, we might have to lower that.





Let's start with the movie metadata.
Where do you get it from?

Extracted from different sources, lots of
duplication, it's a mess now.



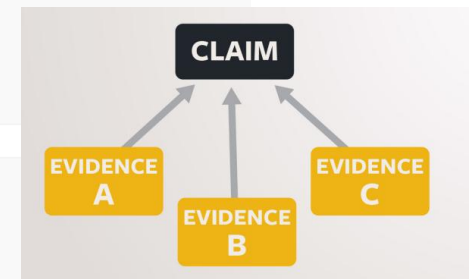
Do you do any deduplication?

Yes, on exact matches for titles and
release_year. But that's too strict, we
miss a lot.



</> JSON

```
{
  "content_id": "M001",
  "title": "The Last Horizon",
  "genre": ["Sci-Fi", "Adventure"],
  "release_year": 2022,
  "actors": [
    {"name": "Emma Stone", "role": "Commander Vale"},
    {"name": "John Boyega", "role": "Pilot Ryn"}
  ],
  "duration_min": 128
}
```



</> JSON

```
{
  "content_id": "M002",
  "title": "The Last Horizon ",
  "genre": ["Science Fiction", "Adventure"],
  "release_year": 2022,
  "actors": [
    {"name": "E. Stone", "role": "Commander Vale"},
    {"name": "John Boyega", "role": "Pilot Ryn"}
  ],
  "duration_min": 127
}
```

```
</> JSON
```

```
{
  "content_id": "M005",
  "title": "Love in Kyoto",
  "genre": ["Romance", "Drama"],
  "release_year": 2020,
  "actors": [
    {"name": "Rinko Kikuchi", "role": "Aiko"},
    {"name": "Henry Golding", "role": "Daniel"}
  ],
  "duration_min": 95
}
```

```
</> JSON
```

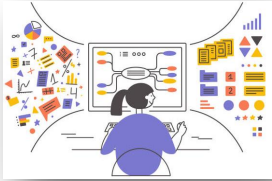
```
{
  "content_id": "M007",
  "title": "The Silent War",
  "genre": ["Drama", "History"],
  "release_year": 2018,
  "actors": [
    {"name": "Mark Rylance", "role": "Thomas Hale"},
    {"name": "Keira Knightley", "role": "Evelyn Price"}
  ],
  "duration_min": 115
}
```

```
</> JSON
```

```
{
  "content_id": "M006",
  "title": "Love in Kyoto (2020)",
  "genre": ["Romance", "Drama"],
  "release_year": 2020,
  "actors": [
    {"name": "Rinko Kikuchi", "role": "Aiko"},
    {"name": "H. Golding", "role": "Daniel"}
  ],
  "duration_min": 96
}
```

```
</> JSON
```

```
{
  "content_id": "M008",
  "title": "The Silent War",
  "genre": ["Action", "Thriller"],
  "release_year": 2023,
  "actors": [
    {"name": "Chris Hemsworth", "role": "Agent Cross"},
    {"name": "Florence Pugh", "role": "Maya Voss"}
  ],
  "duration_min": 130
}
```

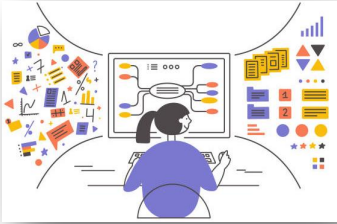


And user information. Do we know a lot about them?



We know this much!

user_id	name	age	gender	neighbourhood	city	signup_date	leave_date
U001	Alex Chen	19	Non-binary	Plateau Mont-Royal	Montreal	2023-01-15	NULL
U002	Maria Gonzalez	52	Female	NULL	Toronto	2022-11-03	2025-06-20
U003	Liam O'Connor	67	NULL	Kitsilano	Vancouver	2021-06-22	NULL
U004	Aisha Khan	31	Female	Scarborough	Toronto	2023-07-10	NULL
U005	Jordan Smith	23	Non-binary	NULL	Ottawa	2024-02-18	NULL
U006	Sophie Tremblay	44	NULL	Rosemont-La Petite-Patrie	Montreal	2023-09-05	2025-01-10
U007	Noah Dubois	36	Male	Griffintown	Montreal	2022-05-27	NULL

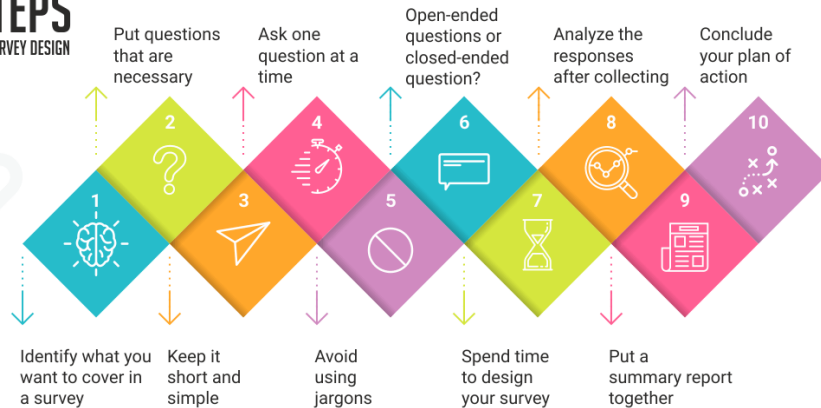


I'm surprised there is no missing values in « Age ».

Trust on the age is really low. It's a mandatory field, but people just enter approximations! Our UI expert says we should use age ranges instead.



10 STEPS TO A GOOD SURVEY DESIGN



QuestionPro

Section 1: Demographics

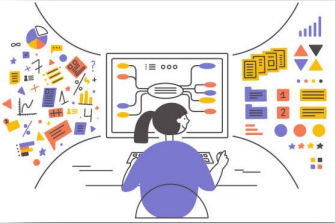
1. Age: _____

2. Gender:

- ☐ Male
- ☐ Female
- ☐ Non-binary / Third gender
- ☐ Prefer not to say

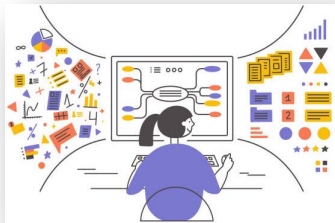
3. Employment/Study Status:

- ☐ Full-time employee
- ☐ Part-time employee
- ☐ Student
- ☐ Other: _____

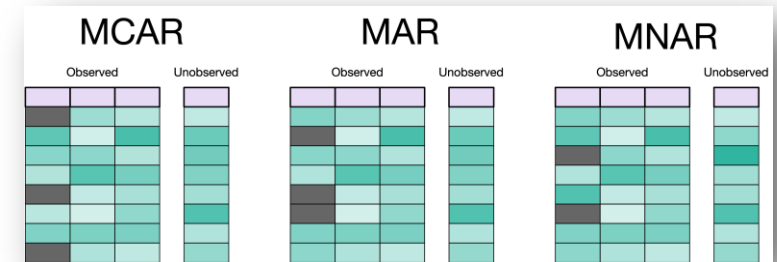


And for gender, it's probably Missing Completely At Random (MCAR).

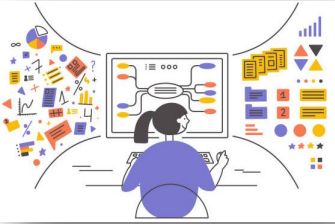
I think so. People just not understanding the relevance.



Not sure if we can do imputation on gender. Perhaps building a gender predictive model based on content watched.

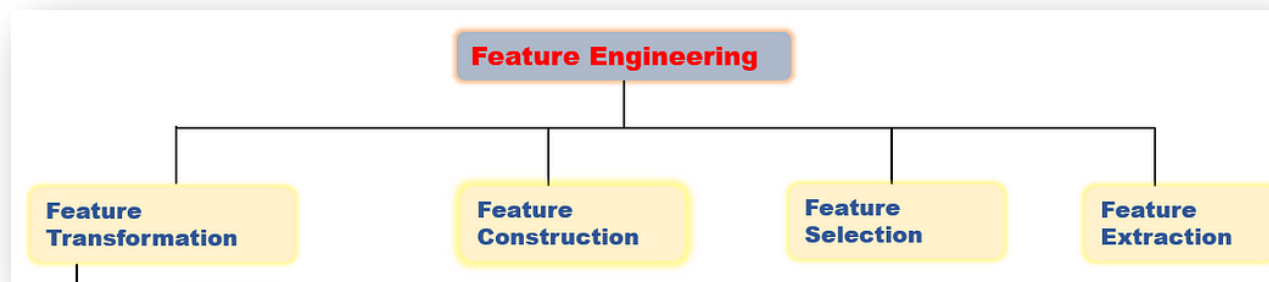


- 1) Arbitrary value imputation
- 2) Default value imputation
- 3) Random Sample imputation
- 4) Time-based imputation
- 5) Conditional imputation
- 6) Correlational imputation
- 7) Regression imputation
- 8) Predictive imputation
- 9) Similarity-based imputation

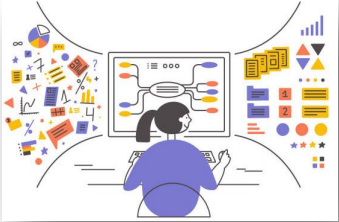


And for spatial aggregation, we'll need to do it at the city level.

Yes, neighbourhood is not mandatory to provide.



Aggregation: statistical, spatial, temporal

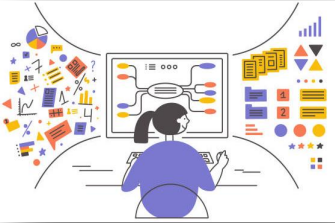


And when people leave and come back, they get the same user_id?



No. Here are some examples with people signing up more than once.

user_id	name	age	gender	neighbourhood	city	signup_date	leave_date
U201	Daniel Reeds	22	Male	Mile End	Montreal	2025-01-08	2025-04-12
U202	Sara Nguyen	41	Female	NULL	Toronto	2025-01-19	2025-03-28
U203	Malik Johnson	36	NULL	Little Italy	Montreal	2025-02-03	2025-06-05
U204	Chloé Martin	29	Female	Plateau Mont Royal	Montreal	2025-02-14	NULL
U205	Ethan Brown	55	Male	Downtown	Vancouver	2025-02-28	2025-06-07
U206	Amina El-Sayed	33	Female	NULL	Ottawa	2025-03-05	2025-04-22
U207	Noah Wilson	27	Male	Scarborough	Toronto	2025-03-18	NULL
U208	Sara Ngyuen	41	Female	Downtown	Toronto	2025-04-02	NULL
U209	Daniel Reed	22	Male	Mile-End	Montreal	2025-04-17	NULL
U210	Amira ElSayed	33	NULL	Little Italy	Montreal	2025-05-01	2025-06-18
U211	Lucas Ferreira	38	Male	NULL	Vancouver	2025-05-12	NULL
U212	Chloe Marten	29	Female	Plateau Mont-Royal	Montreal	2025-05-28	NULL
U213	Ethan Browne	55	Male	Downtown	Vancouver	2025-06-10	NULL
U214	Malik Jonson	36	Male	Little Italy	Montreal	2025-06-25	NULL
U215	Amina El Sayed	33	Female	Ottawa	Ottawa	2025-07-04	NULL

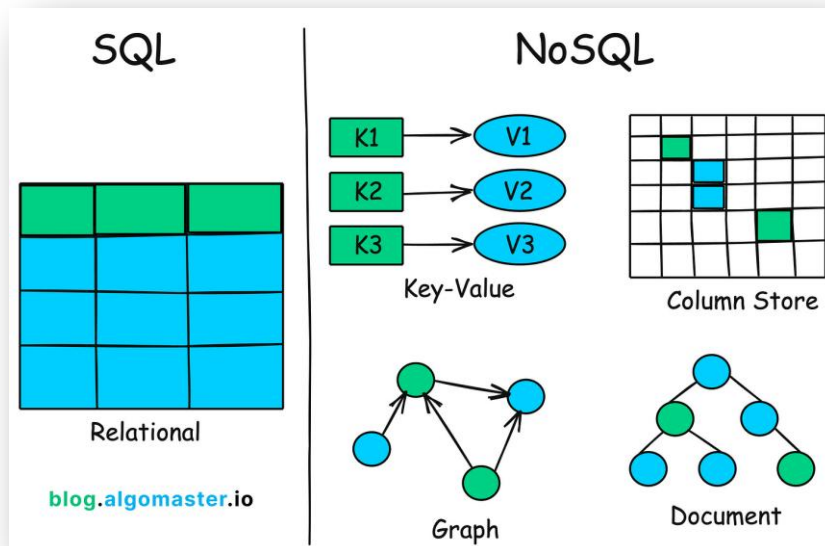


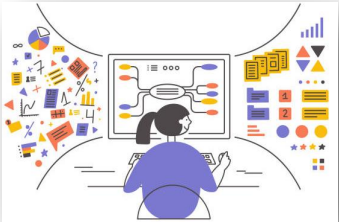
Maybe a different type of storage would be appropriate.

Yes, we might be better off with a document database. More flexible.



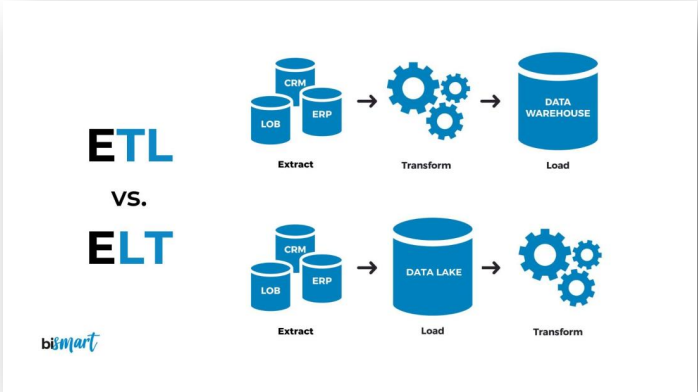
```
{
  "user_id": "P002",
  "canonical_name": "Sara Nguyen",
  "age": 41,
  "gender": "Female",
  "accounts": [
    {
      "user_id": "U202",
      "name": "Sara Nguyen",
      "signup_date": "2025-01-19",
      "leave_date": "2025-03-28",
      "neighbourhood": null,
      "city": "Toronto"
    },
    {
      "user_id": "U208",
      "name": "Sara Ngyuen",
      "signup_date": "2025-04-02",
      "leave_date": null,
      "neighbourhood": "Downtown",
      "city": "Toronto"
    }
  ]
}
```





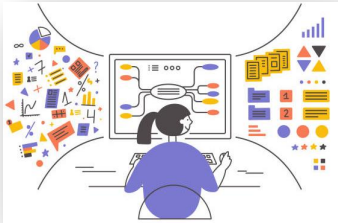
And you say you have viewing logs. Those are in the data lake?

So far, yes. But if you have a fixed processing you need, I can build an ETL pipeline to get you the data in better shape for analysis.



event_id	user_id	content_id	event_type	timestamp
E001	U001	M001	play	2024-03-01 19:02:15
E002	U001	M001	stop	2024-03-01 20:00:00
E003	U001	M001	play	2024-03-02 21:10:00
E004	U001	M001	stop	2024-03-02 21:55:00
E005	U001	M001	play	2024-03-03 18:20:10
E006	U001	M001	stop	2024-03-03 18:42:30
E007	U002	M002	play	2024-03-02 20:15:05
E008	U002	M002	stop	2024-03-02 21:50:20
E009	U001	M003	play	2024-03-05 18:10:44
E010	U001	M003	stop	2024-03-05 19:10:10
E011	U003	M001	play	2024-03-06 20:05:33
E012	U003	M001	stop	2024-03-06 22:13:45
E013	U004	M002	play	2024-03-07 21:00:12
E014	U004	M002	stop	2024-03-07 22:50:30
E015	U005	M003	play	2024-03-08 17:25:55
E016	U005	M003	stop	2024-03-08 18:55:10

event_id	user_id	content_id	event_type	timestamp
E001	U001	M001	play	2024-03-01 19:02:15
E002	U001	M001	stop	2024-03-01 20:00:00
E003	U001	M001	play	2024-03-02 21:10:00
E004	U001	M001	stop	2024-03-02 21:55:00
E005	U001	M001	play	2024-03-03 18:20:10
E006	U001	M001	stop	2024-03-03 18:42:30
E007	U002	M002	play	2024-03-02 20:15:05
E008	U002	M002	stop	2024-03-02 21:50:20
E009	U001	M003	play	2024-03-05 18:10:44
E010	U001	M003	stop	2024-03-05 19:10:10
E011	U003	M001	play	2024-03-06 20:05:33
E012	U003	M001	stop	2024-03-06 22:13:45
E013	U004	M002	play	2024-03-07 21:00:12
E014	U004	M002	stop	2024-03-07 22:50:30
E015	U005	M003	play	2024-03-08 17:25:55
E016	U005	M003	stop	2024-03-08 18:55:10



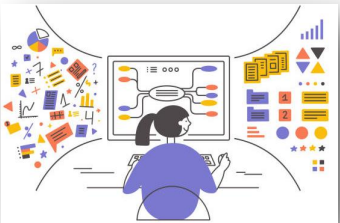
For sure I would like to derive duration,
no need for the raw start/stop.

user_id	content_id	total_duration_min
U001	M001	162
U002	M002	95
U001	M003	59
U003	M001	128
U004	M002	110
U005	M003	89



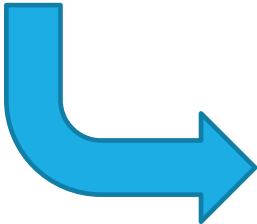
Sounds good!

event_id	user_id	content_id	event_type	timestamp
E001	U001	M001	play	2024-03-01 19:02:15
E002	U001	M001	stop	2024-03-01 20:00:00
E003	U001	M001	play	2024-03-02 21:10:00
E004	U001	M001	stop	2024-03-02 21:55:00
E005	U001	M001	play	2024-03-03 18:20:10
E006	U001	M001	stop	2024-03-03 18:42:30
E007	U002	M002	play	2024-03-02 20:15:05
E008	U002	M002	stop	2024-03-02 21:50:20
E009	U001	M003	play	2024-03-05 18:10:44
E010	U001	M003	stop	2024-03-05 19:10:10
E011	U003	M001	play	2024-03-06 20:05:33
E012	U003	M001	stop	2024-03-06 22:13:45
E013	U004	M002	play	2024-03-07 21:00:12
E014	U004	M002	stop	2024-03-07 22:50:30
E015	U005	M003	play	2024-03-08 17:25:55
E016	U005	M003	stop	2024-03-08 18:55:10

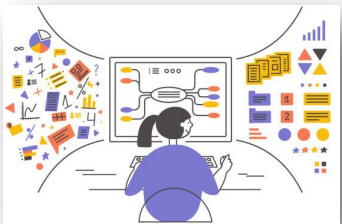


Oh might as well keep the first and last timestamp too.

user_id	content_id	total_duration_min	first_play_timestamp	last_stop_timestamp
U001	M001	162	2024-03-01 19:02:15	2024-03-03 18:42:30
U002	M002	95	2024-03-02 20:15:05	2024-03-02 21:50:20
U001	M003	59	2024-03-05 18:10:44	2024-03-05 19:10:10
U003	M001	128	2024-03-06 20:05:33	2024-03-06 22:13:45
U004	M002	110	2024-03-07 21:00:12	2024-03-07 22:50:30
U005	M003	89	2024-03-08 17:25:55	2024-03-08 18:55:10



OK. I can also further derive the number of days across which they watched the movie.



Where do you keep the ratings?
You haven't mentioned them so far.

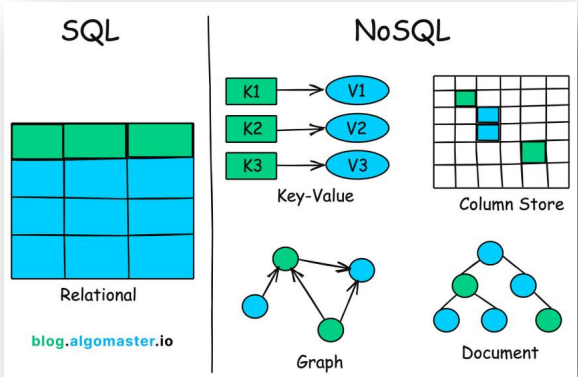
We keep them in a key value store
and use them for popularity sorting.

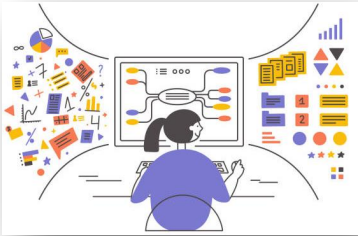


Key	Value
U001:M001	{ "rating": 5, "timestamp": "2024-03-01" }
U002:M002	{ "rating": 4, "timestamp": "2024-03-02" }
U001:M003	{ "rating": 3, "timestamp": "2024-03-05" }
U003:M001	{ "rating": 5, "timestamp": "2024-03-06" }
U004:M002	{ "rating": 4, "timestamp": "2024-03-07" }
U005:M003	{ "rating": 4, "timestamp": "2024-03-08" }

Aggregate once a day

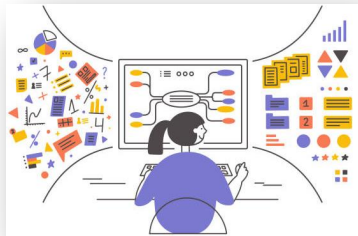
Rank	content_id	avg_rating	rating_count	popularity_score
1	M003	3.67	3	11
2	M001	5.00	2	10
3	M002	4.00	2	8



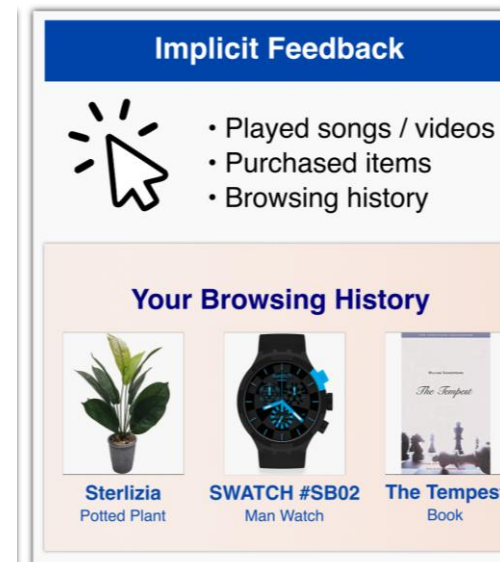
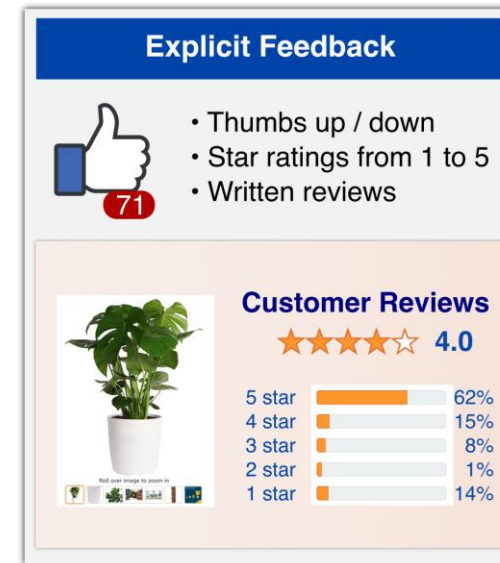


They are not used to make recommendations?

No because most users do not rate anything.



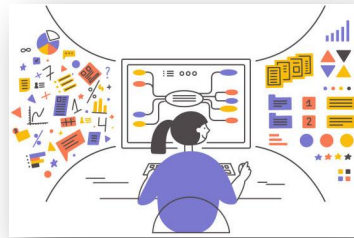
Perhaps we try implicit feedback. Just derive a like/notlike from watch time and movie duration.





Then we need to also derive a like/not like from the 5* rating.

Maybe change the UI so people would do thumbs up/down instead of rating.

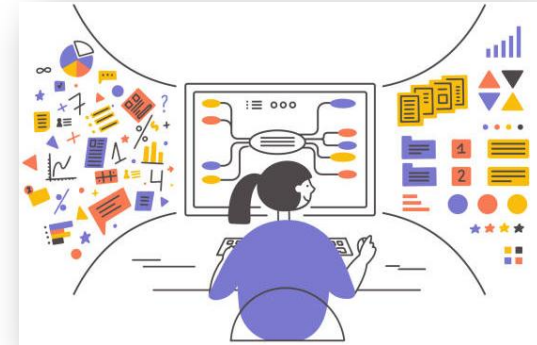


You should talk to the UI expert.

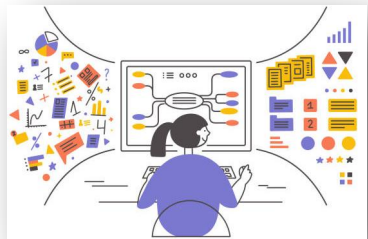
Easy to get carried away BECAUSE there are so many things we could do and would like to do.

The manager *(again)*



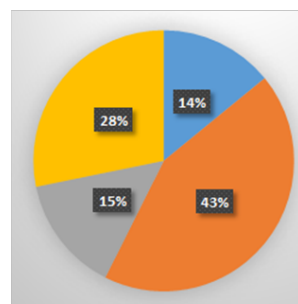


We hired you to figure out why users are leaving the platform.
We want you to help us find out more about our viewers.



I've looked at many variables independently, did some correlation and also did some clustering.

Watch time

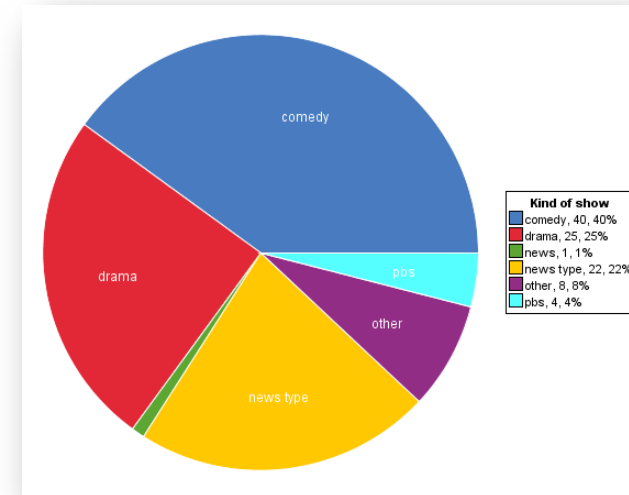


0-10 hours per week

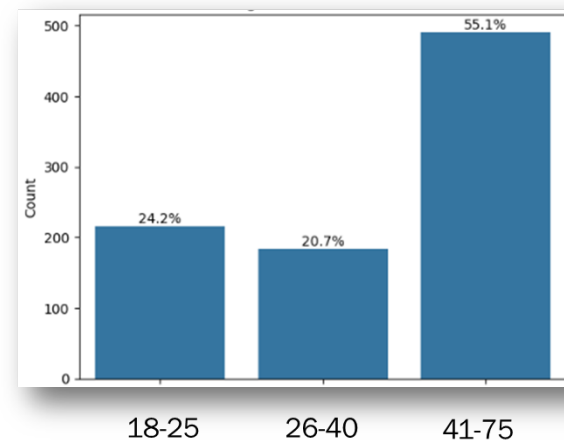
11-20 hours per week

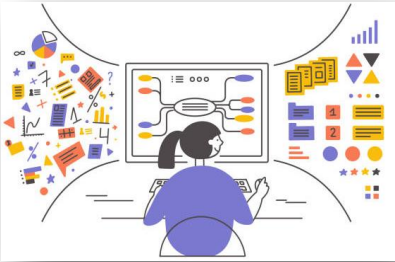
21-30 hours per week

31 and more hours per week

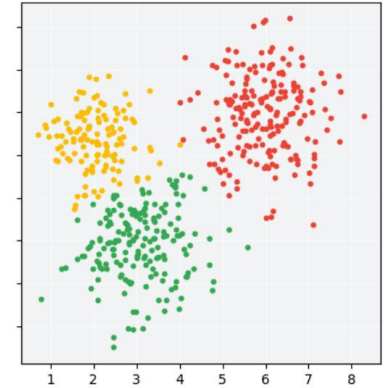


Churn rate (after the one-month free trial)





The young sci-fi lovers have 15% churn rate.
 The midrange romance lovers have 5% churn rate.
 The « explorers » (mixed genres, no ratings) have 40% churn rate.
 The high raters have 10% churn rate.
 The « series/film fans » (over 30 hours per week) have 2% churn rate.

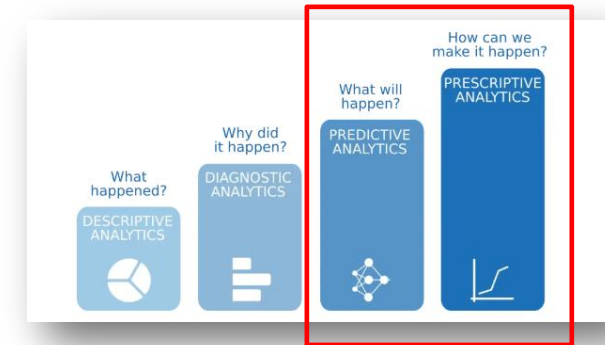


We should then:

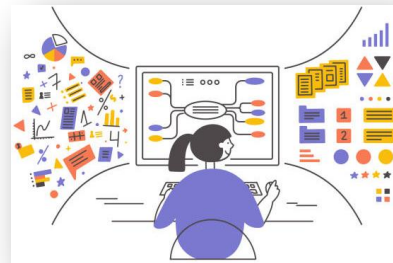
- Have a **preventive approach**, to try to increase watch time, provide better recommendations and encourage rating.
- Have a **corrective approach** (retention offer) with 50% off for staying another month.



Can you provide some ideas on the implementation of the preventive approach and the corrective one.



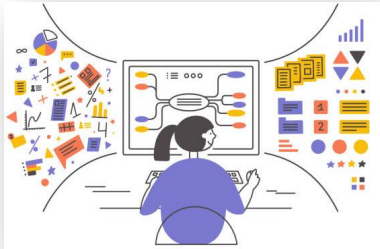
Sure. And we can continue learning about our users. I would suggest to do a survey to learn about the viewers. We could learn a lot from their responses.



We cannot keep emails of previous users, so we would only burden the users staying.

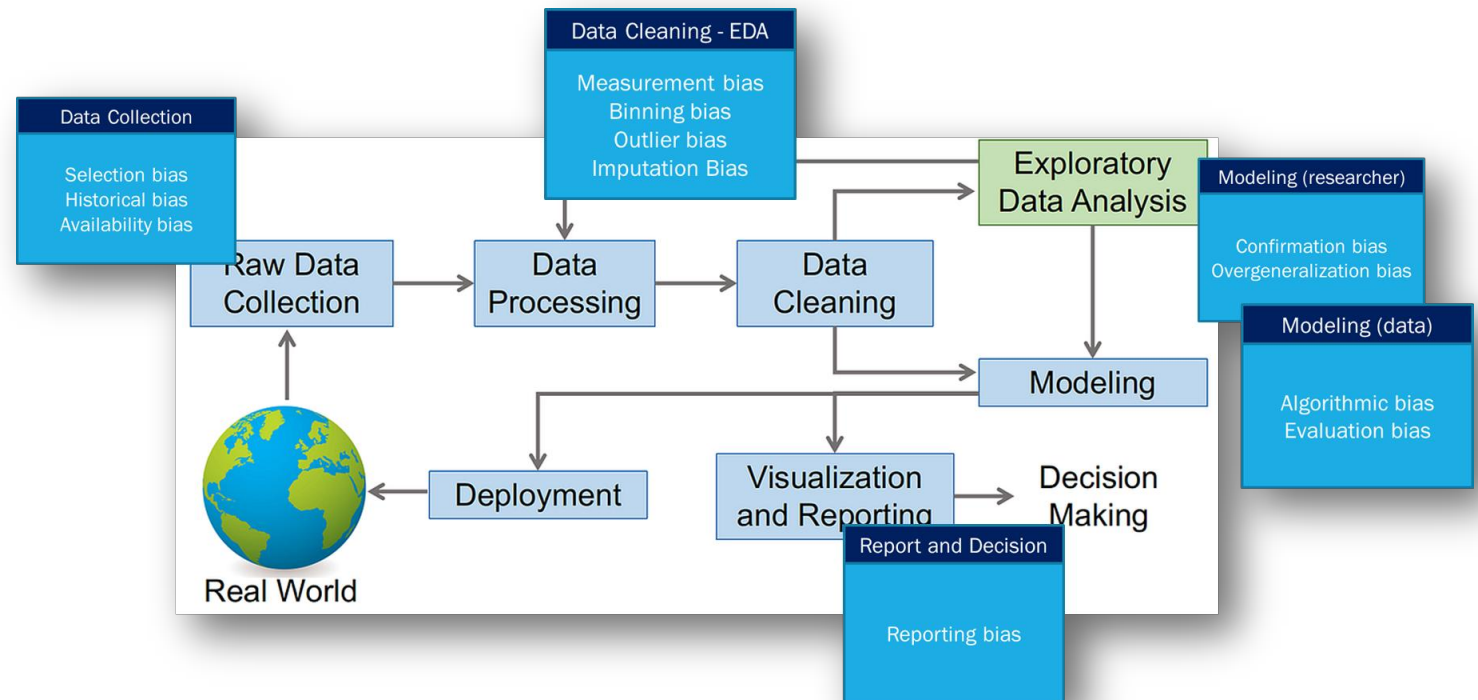


Self-Reported Data	Surveys Interviews Focus groups
Data directly provided by individuals.	
Automated Data	Logs Sensor Transactions
Data generated obtained through an automated system.	

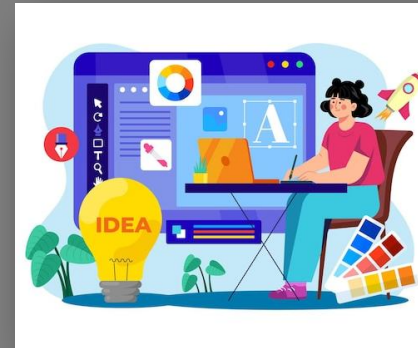


I see. Even if we asked questions about what they like or not like, our survey would have a convergence bias. But perhaps something short (e.g. tell us what makes you leave...) would still be informative.

We don't want to annoy them... The UI designer might know a good way to do that.

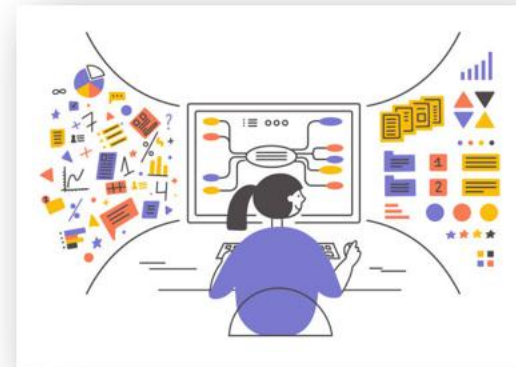
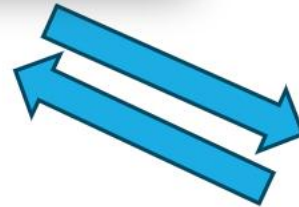


The UI designer

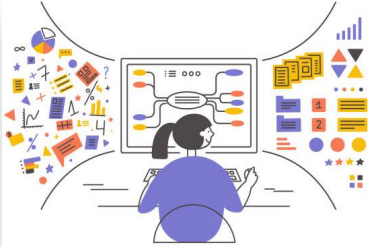


UI Designer

- User experience
- Interpretability
- Cognitive load
- Interaction design



Data Scientist

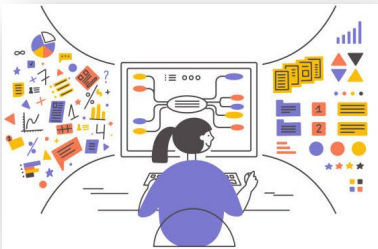


I'm hired to understand, prevent and/or correct churn. What about a small survey when people are about to leave?

You don't want to annoy them. But sure, that's easy to do, and we need to well design the one-question so the answers become actionable items.



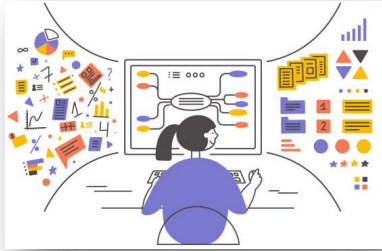
Absolutely! I've already done an analysis that can guide our choices (e.g. not enough content of the genre you like, poor recommendations, etc)



We are sad to see you leave us.

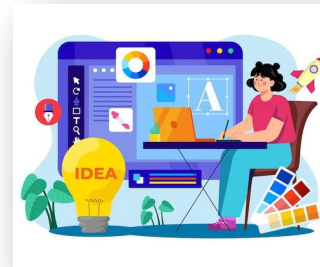
Please tell us why you decided to cancel

- ☒ Too expensive
- ☐ Missing key features
- ☐ Poor customer support
- ☐ Found a better alternative
- ☐ Other (please specify)



Also, what is your opinion on a new rating system (thumbs up/down) instead of 1 to 5 stars.

There is HCI research showing it's better as it has less cognitive load on the users. You should get more people rating.



Designing the user experience of a rating system

In designing a new product, I needed a scoring system. Users should vote for a series of parameters, and the first thing that comes to mind is the legacy 5-star rating system. But in recent years this rating system has shown that it's not optimized and/or efficient.

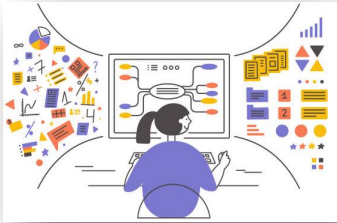


Hossein Sharafi

Follow

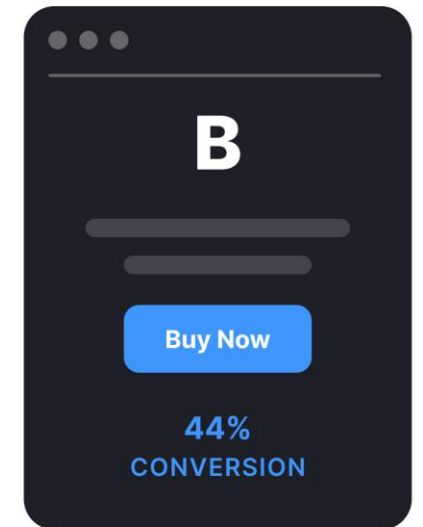
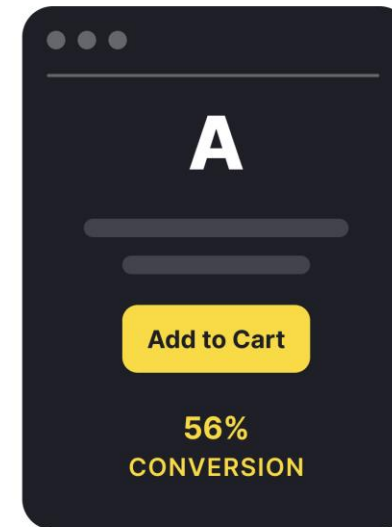
6 min read · Mar 7, 2022

[Source](#)

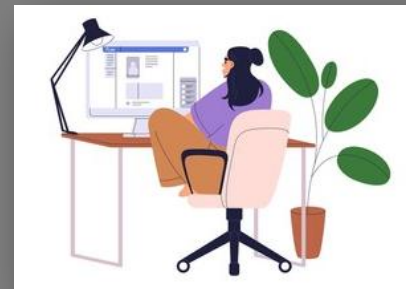


Great! Can we implement it within an A/B testing approach?

Not an issue for me, I can provide a few design options to test.



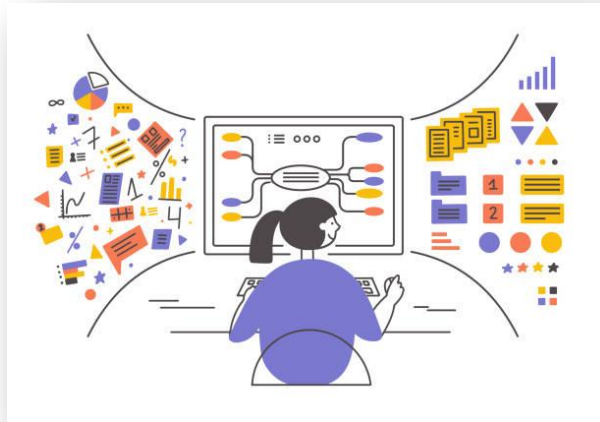
The ML expert



Machine Learning Expert

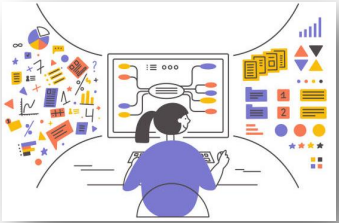


- Model choice
- Model performance
- Underlying assumptions
- Mathematical rigor
- Proper evaluation



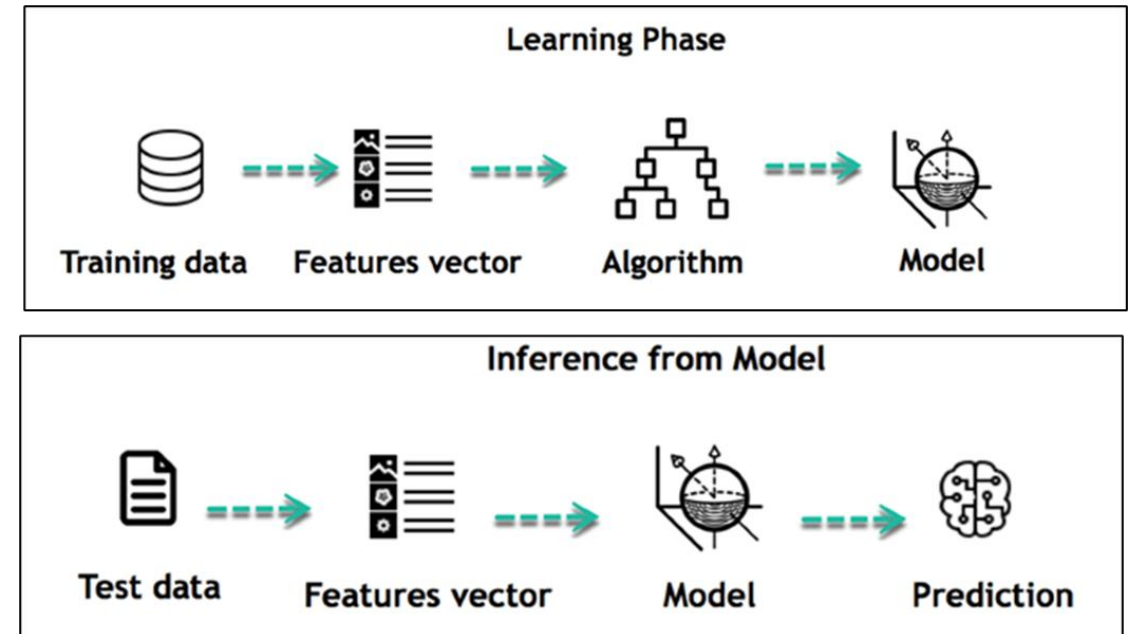
Data Scientist





I'm looking into building a predictive model of customers likely to leave the platform.

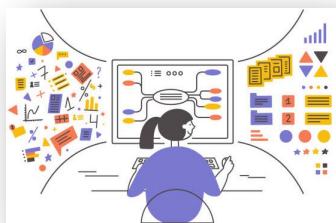
Sure, let's brainstorm about the task, data, models and evaluation.





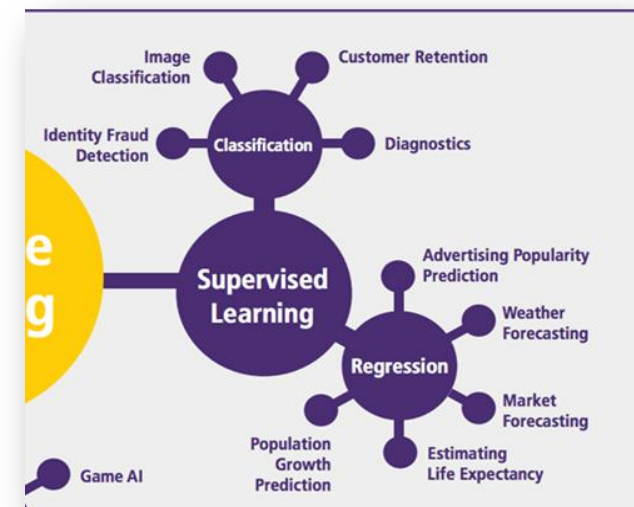
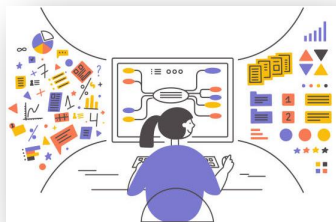
How do you see the task?

As a binary classification



Why not a regression?

Not sure we need such granularity. Or multiclass?

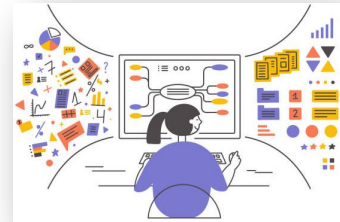


Movie	Features			Possible output	
	Duration	Producer	Lead actor/actress	<u>Classification</u> Genre	<u>Regression</u> Gross Revenue
M1	90	P1	A1	Drama	3.2M
M2	140	P3	A2	Drama	200K
M3	80	P1	A1	Comedy	15M
M4	90	P1	A3	Drama	500K
M5	100	P3	A4	Drama	10K
M6	110	P2	A5	Comedy	1.3M
M7	100	P2	A5	Comedy	870K



What data do you have?

Lots of data gathered in data lakes about users, movies and viewing.



Oh... lots of aggregation will be needed. Let's filter and organize.

User Data

- UserID
- Age
- Gender
- Neighbourhood
- City
- Signup date
- Leave date

Content Data

- MovieID
- Movie title
- Genres
- Actors
- Duration
- Release Year

Viewing data

- UserID
- MovieID
- Duration
- First play
- Last Stop
- Rating (explicit or implicit)



Demographics

```
{  
  "user_id": "P001",  
  "is_active": true,  
  "profile": {  
    "age": 22,  
    "gender": "Male",  
    "city": "Montreal",  
    "neighbourhood": "Mile End"  
  },  
}
```

Platform access

```
"lifecycle": {  
  "account_age_days": 90,  
  "num_reactivations": 1,  
  "total_lifetime_days": 150  
},
```

Engagement

```
"engagement": {  
  "total_watch_time_min": 540,  
  "num_sessions": 12,  
  "avg_session_duration": 45,  
  "last_activity": "2025-07-08"  
},
```



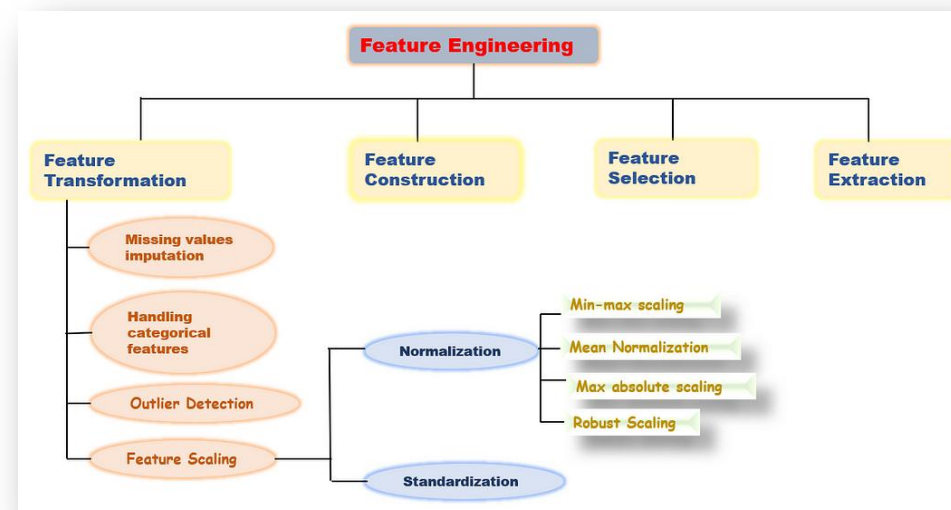


Ratings

```
"ratings": {
  "avg_rating": 4.2,
  "num_ratings": 10
},
```

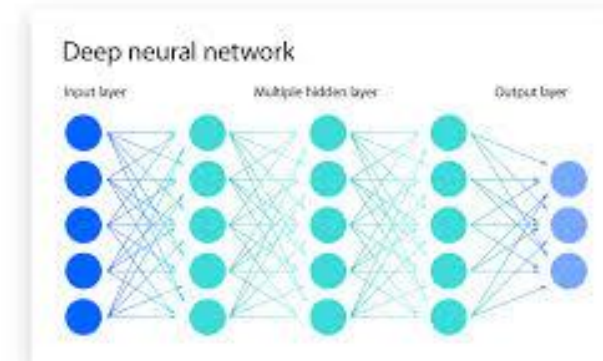
Preferences

```
"preferences": {
  "top_genre": "Sci-Fi",
  "genre_distribution": {
    "Sci-Fi": 0.5,
    "Drama": 0.3,
    "Crime": 0.2
  }
},
```

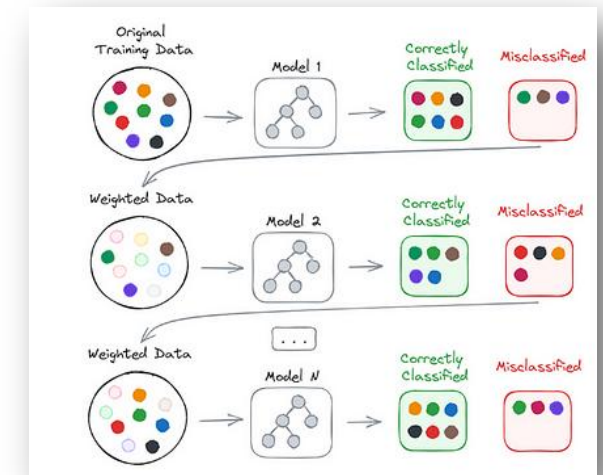
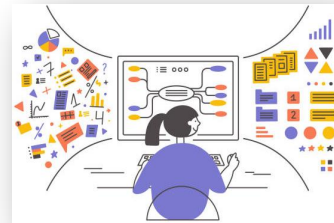




If you have tons of data, you could use a neural network (MLP).



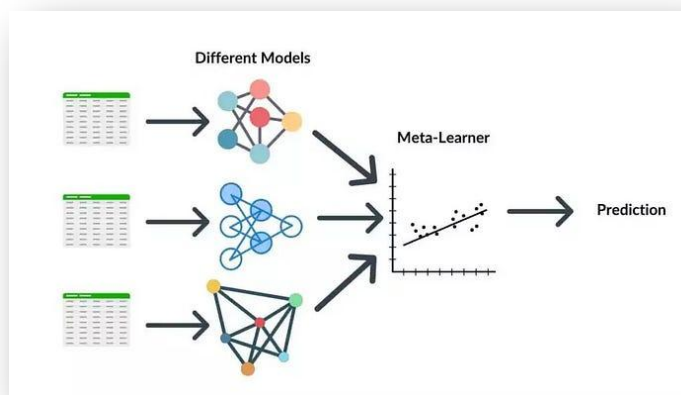
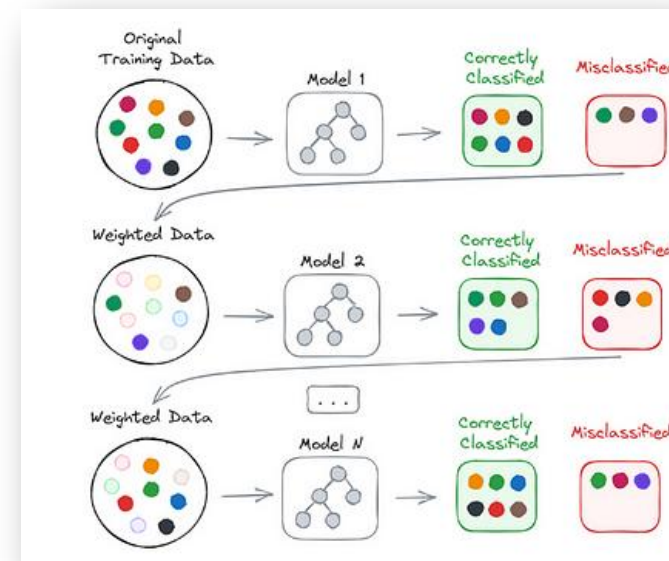
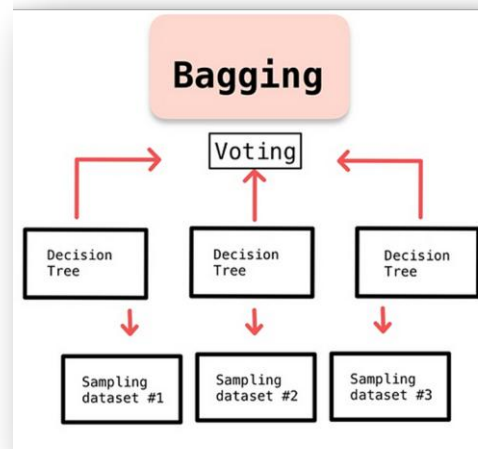
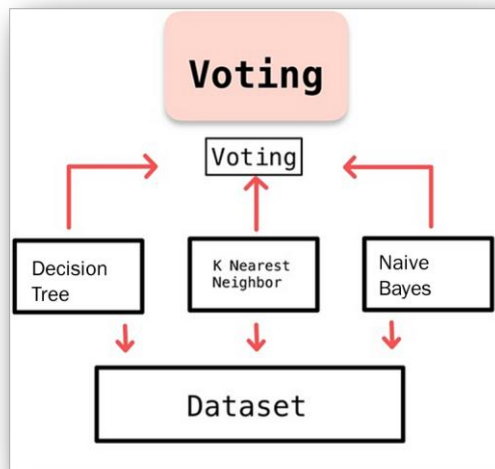
Would be good to have explainability...
Although, if we get better churn prediction allowing us to better target retention strategies, then maybe explainability is not necessary for action, but would be interesting for understanding our customers.



You can do both, like a decision tree (maybe adaboost) and a MLP and compare performances.

$$H_{\text{final}} = \text{sign} \left[0.42 \cdot \begin{array}{|c|c|c|} \hline + & - & - \\ \hline + & - & - \\ \hline \end{array} + 0.65 \cdot \begin{array}{|c|c|c|} \hline + & - & - \\ \hline + & - & - \\ \hline \end{array} + 0.92 \cdot \begin{array}{|c|c|c|} \hline + & - & - \\ \hline + & - & - \\ \hline \end{array} \right]$$

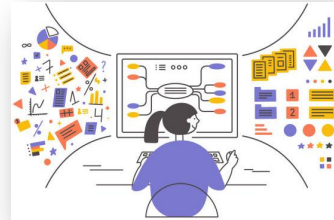
$$= \begin{array}{|c|c|c|} \hline + & - & - \\ \hline + & - & - \\ \hline \end{array}$$



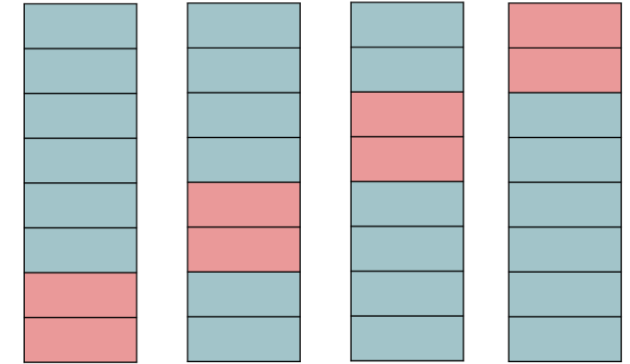


How to evaluate?

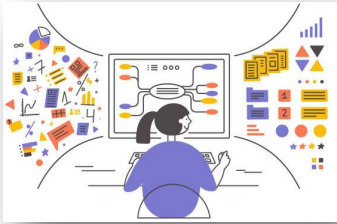
Do cross-validation.
Build a balanced training set with
churn/not churn. But what about
reactivation?



4-fold cross-validation



Maybe do multi-class (churn in first month, churn within 6 months, churn-return, no churn). At this point, best to go back to the manager to help define these categories. What exactly would he like to predict?



Yes, I'll go to validate the outputs of the predictive model.



Once that is decided, go see the Data Engineer so he builds a pipeline to extract all that information for you.

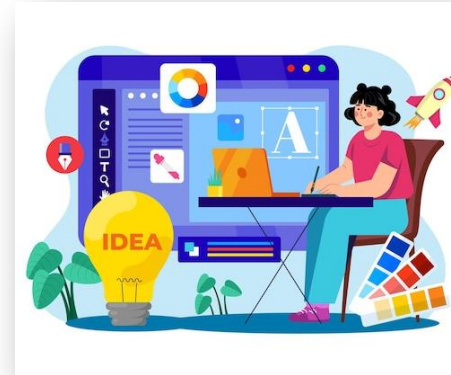


Summary

- Data ingestion pipelines
- Real-time versus batch processing
- Data consistency
- Storage architecture



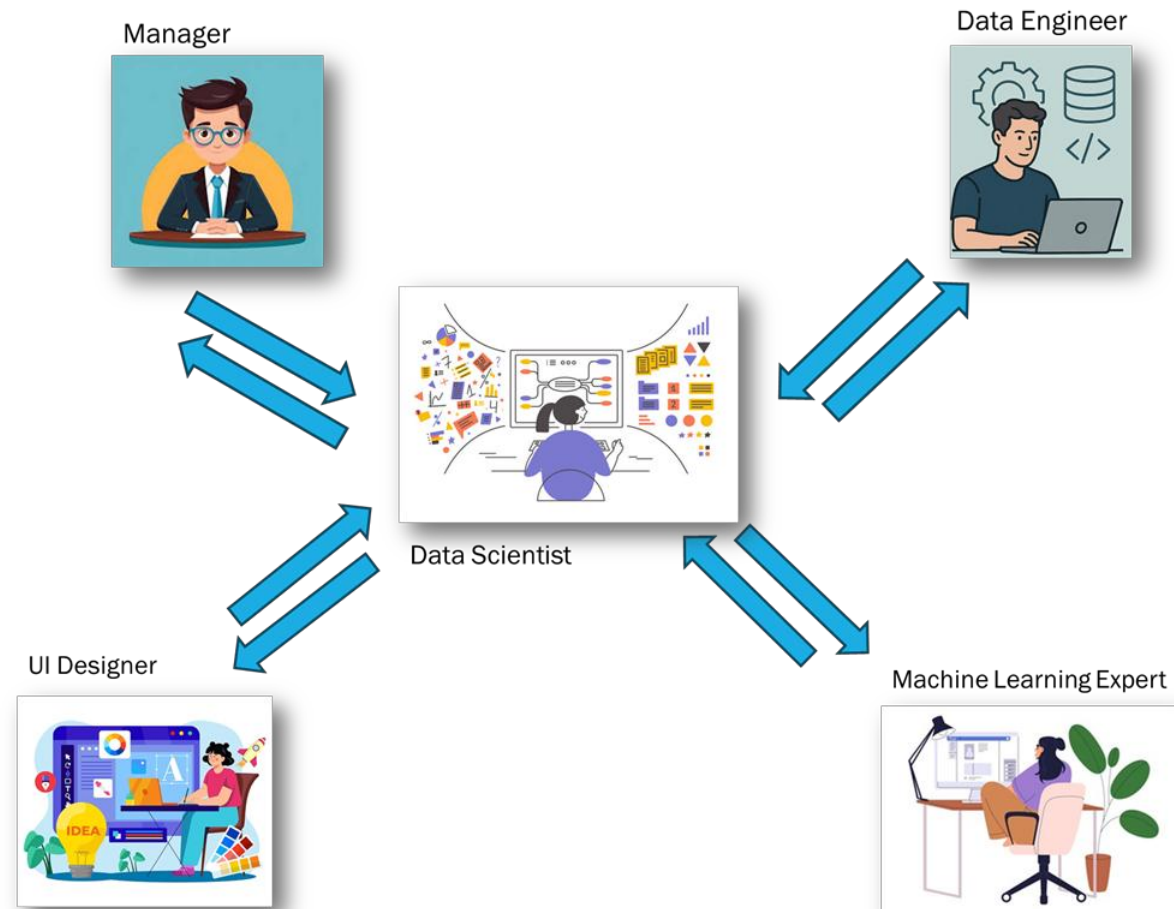
- Subscriber growth and retention
- Watch time
- Revenue impact
- Clear, actionable insights

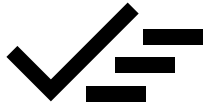


- User experience
- Interpretability
- Cognitive load
- Interaction design



A data scientist job is to well understand the data process and also navigate competing constraints across people, systems and goals.





DATA SCIENCE PROCESS

- Introduction
- Case Study – Streaming Platform
 - Interaction with the manager
 - Interaction with the data engineer
 - Interaction with the UI designer
 - Interaction with the ML expert
- Summary